Mirror Descent: non Euclidean geometries

$$x_{t+1} = \underset{x \in \mathcal{X}}{\operatorname{argmin}} \; \eta g_t^T x + \underbrace{D_\phi(x, x_t)}$$

$\frac{1}{2}\|x - x_t\|_2^2$ – recovers standard (proj) sub-grad descent.

Key constants in rate of conv: L - Lipschitz.

Lipschitz, smooth and strongly convex: different norms

Lipschitz

$f : \mathbb{R}^n \to \mathbb{R}$

$|f(x) - f(y)| \le L \cdot \|x - y\|$ ← arbitrary norm.

If f is convex: $f(y) \ge f(x) + \nabla f(x)^T (y - x)$

$f(x) - f(y) \le \boxed{\nabla f(x)^T (x - y)}$

$\Rightarrow |f(x) - f(y)| \le \|\nabla f(x)\|_* \cdot \|x - y\|$

Another definition: f is L-Lipschitz w.r.t. $\|\cdot\|$

if $\|\nabla f(x)\|_* \le L$.

Lipschitz, smooth and strongly convex: different norms

Smooth 1/2

arbitrary norm

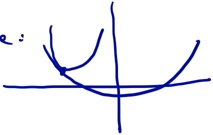
f is β-smooth w.r.t. $\|\cdot\|$ if

$$\boxed{\|\nabla f(x) - \nabla f(y)\|_* \le \beta \cdot \|x-y\|}$$

Recall that for our $\|\cdot\|_2$-definition of β-smoothness, we showed:

$$f(y) \le f(x) + \nabla f(x)^T (y-x) + \frac{\beta}{2}\|x-y\|_2^2 .$$

Picture:



Want to show that for our general-norm definition of smoothness, something similar holds.

Remember the FTC:
$$\int_0^1 \left(\frac{d}{dt} F(t)\right) dt = F(1) - F(0).$$

Lipschitz, smooth and strongly convex: different norms

Want: $f(x) \le f(y) + \nabla f(y)^T(x-y) + \frac{\beta}{2}\|x-y\|^2$ $\circledast$

Smooth 2/2

$$\boxed{|f(x) - f(y) - \nabla f(y)^T(x-y)|} = \left| \int_0^1 \Big(\underbrace{\frac{d}{dt}\big(f(y + t(x-y))\big)}_{\text{need chain rule}} - \nabla f(y)^T(x-y) \Big)\, dt \right|$$

$$= \left| \int_0^1 \left(\nabla f(y + t(x-y))^T (x-y) - \nabla f(y)^T(x-y) \right) dt \right|$$

$$= \left| \int_0^1 \left(\nabla f(y + t(x-y)) - \nabla f(y) \right)^T (x-y)\, dt \right|$$

$$(u^T v \le \|u\|_* \cdot \|v\|)$$

$$\le \int_0^1 \underbrace{\| \nabla f(y + t(x-y)) - \nabla f(y) \|_*} \cdot \|x-y\|\, dt$$

$$\le \int_0^1 \beta \cdot t \cdot \|x-y\| \cdot \|x-y\|\, dt = \beta \cdot \|x-y\|^2 \cdot \int_0^1 t\, dt$$

$$= \boxed{\frac{\beta}{2}\|x-y\|^2}.$$

Lipschitz, smooth and strongly convex: different norms

Strongly convex

f is α - strongly convex w.r.t. $\|\cdot\|$:

$$f(y) \geq f(x) + g^T(y-x) + \frac{\alpha}{2}\|x-y\|^2 .$$

$g \in \partial f(x)$

Mirror Descent: the result

$\min: f(x)$ Let f be L-Lipschitz w.r.t. $\|\cdot\|$

$\text{st}: x \in \mathcal{X}$ $\phi: \mathcal{D} \to \mathbb{R}$, $\mathcal{X} \subseteq \bar{\mathcal{D}}$, $\mathcal{X} \cap \mathcal{D} \neq \phi$

ϕ is ρ-strongly convex w.r.t. $\|\cdot\|$ $\Downarrow$

Recall: $D_\phi(x,y) = \phi(x) - (\phi(y) + \nabla\phi(y)^T(x-y)) \geq \frac{\rho}{2}\|x-y\|^2$.

Also assume: $\nabla\phi: \mathcal{D} \to \mathbb{R}^n$, $\lim_{x \to \partial\mathcal{D}} \|\nabla\phi(x)\|_* = +\infty$.

Algorithm (MD): $x_{t+1} = \underset{x \in \mathcal{X}}{\operatorname{argmin}}: \eta g_t^T x + D_\phi(x, x_t)$

$$\eta = \frac{R}{L} \cdot \sqrt{\frac{2\rho}{T}}, \quad R^2 \doteq D_\phi(x_1, x^*)$$

Thm: $f\left(\frac{1}{T}\sum_{t=1}^{T} x_t\right) - f(x^*) \leq RL \cdot \sqrt{\frac{2}{\rho T}}$

Observations: ① $\frac{1}{\sqrt{T}}$ – same dependence on T. ② L is w.r.t. $\|\cdot\|$, not $\|\cdot\|_2$.

Mirror descent - proof

$$0 \leq \eta \left(f(x_t) - f(x^*)\right)$$

$$\leq \eta \cdot g_t^T (x_t - x^*) \qquad \left(\text{by convexity of } f \text{ — } g_t \in \partial f(x_t)\right)$$

$$= \left(\nabla\phi(x_t) - \nabla\phi(x_{t+1}) - \eta g_t\right)^T (x^* - x_{t+1}) \qquad S1$$

$$+ \left(\nabla\phi(x_{t+1}) - \nabla\phi(x_t)\right)^T (x^* - x_{t+1}) \qquad S2$$

$$+ \eta g_t^T (x_t - x_{t+1}) \qquad S3$$

$$\leq S2 + S3$$

We will now bound each of S1, S2, S3.

Mirror descent - proof

S_1 $\quad (\nabla\phi(x_t) - \nabla\phi(x_{t+1}) - \eta g_t)^T (x^* - x_{t+1})$

$$x_{t+1} = \arg\min_{x\in\mathcal{X}} : \ \eta g_t^T x + \cancel{D_\phi(x, x_t)} \ \phi(x) - \phi(x_t) - \nabla\phi(x_t)^T(x - x_t)$$

$$0 \in \eta g_t + \nabla\phi(x_{t+1}) - \nabla\phi(x_t) + N_{\mathcal{X}}(x_{t+1})$$

$$\eta g_t + \nabla\phi(x_{t+1}) - \nabla\phi(x_t) \in -N_{\mathcal{X}}(x_{t+1})$$

$$\iff \quad (\eta g_t + \nabla\phi(x_{t+1}) - \nabla\phi(x_t))^T (x - x_{t+1}) \geq 0 \quad \forall x \in \mathcal{X}$$

Since it holds for <u>all</u> $x \in \mathcal{X}$, it also holds for x^*

plugging in x^* for x, we have:

$$\boxed{S_1 \leq 0}$$

Mirror descent - proof

S2 :

$$(\nabla\phi(x_{t+1}) - \nabla\phi(x_t))^T (x^* - x_{t+1})$$

$$= -\left(D_\phi(x_{t+1}, x_t) + D_\phi(x^*, x_{t+1}) - D_\phi(x^*, x_t)\right)$$

(simple calculation & definition of D_ϕ)

$$= D_\phi(x^*, x_t) - D_\phi(x_{t+1}, x_t) - D_\phi(x^*, x_{t+1})$$

S3 : Use : $u^T v \leq \frac{1}{2\alpha} \|u\|_*^2 + \frac{\alpha}{2} \|v\|^2$

$$\underbrace{\eta g_t^T}_{u} \underbrace{(x_t - x_{t+1})}_{v} \leq \frac{1}{2\rho} \cdot \eta^2 \cdot \|g_t\|_*^2 + \frac{\rho}{2} \|x_t - x_{t+1}\|^2 .$$

Mirror descent - proof

We have: $0 \le \eta\,(f(x_t) - f(x^*)) \le s1 + s2 + s3 \le s2 + s3$

$$\le \underbrace{D_\phi(x^*, x_t) - D_\phi(x^*, x_{t+1})}_{\text{telescoping}} \underbrace{- D_\phi(x_{t+1}, x_t) + \frac{\rho}{2}\|x_t - x_{t+1}\|^2} + \frac{1}{2\rho}\eta^2\|g_t\|_*^2$$

$$\le \underline{D_\phi(x^*, x_t) - D_\phi(x^*, x_{t+1})} + \frac{1}{2\rho}\eta^2\|g_t\|_*^2$$

$$\left(\text{b/c} \quad D_\phi(x_{t+1}, x_t) \ge \underline{\frac{\rho}{2}\|x_{t+1} - x_t\|^2}\right)$$

Sum both sides $t = 1$ to T:

$$\eta \cdot \sum_{t=1}^{T}(f(x_t) - f(x^*)) \le D_\phi(x^*, x_1) - \cancel{D_\phi(x^*, x_{T+1})} + \frac{\eta^2}{2\rho}\sum \underbrace{\|g_t\|_*^2}_{L^2}$$

$$\Rightarrow \boxed{f\left(\frac{1}{T}\sum_{t=1}^{T} x_t\right) - f(x^*) \le \frac{\overset{=R^2}{D_\phi(x^*, x_1)}}{\eta T} + \frac{\eta}{2\rho}\cdot L^2}$$

Mirror descent - proof

Finally: optimize choice of η :

$$\frac{R^2}{\eta T} + \frac{\eta}{2\rho} \cdot L^2 \quad \text{convex in } \eta.$$

$$\frac{d}{d\eta}: \quad -\frac{R^2}{T} \cdot \frac{1}{\eta^2} + \frac{L^2}{2\rho} = 0$$

$$\eta = \frac{R}{L} \cdot \sqrt{\frac{2\rho}{T}}$$

Plug this in to the above bound:

$$f\left(\frac{1}{T} \sum x_t\right) - f(x^*) \leq RL \cdot \sqrt{\frac{2}{\rho T}}.$$

Mirror descent - proof

Mirror descent - proof